

PETROLINA, Brazil

WMO: 829840

Lat: 9.362S

Lon: 40.569W

Elev: 385

StdP: 96.78

Time Zone: -3.00 (W03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	19.9	8.1	7.0	32.1	9.2	7.6	31.7	9.2	25.7	8.5	26.0	3.1	140	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.4	37.2	21.5	36.2	21.3	35.9	21.3	24.9	29.9	24.5	29.8	24.0	29.6	3.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.7	25.9	23.1	18.8	25.9	22.8	18.4	25.9	77.9	30.2	76.0	29.2	74.4	29.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	7.0	DB	16.7	39.0	1.3	0.8	15.7	39.6	15.0	40.1	14.2	40.5	13.2	41.1	(4)
(5)			WB	13.7	25.9	1.1	1.0	12.9	26.6	12.2	27.1	11.6	27.7	10.8	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	29.0	28.9	29.3	28.5	27.5	26.3	25.5	26.0	27.3	28.9	29.7	29.3	(6)	
	DBStd	1.93	1.28	1.39	1.27	1.46	1.26	1.23	1.18	1.16	1.53	1.21	1.41	1.64	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6574	589	530	598	554	543	489	482	496	519	585	591	598	(10)	
	CDD18.3	3532	331	297	340	304	285	239	223	238	269	327	341	339	(11)	
	CDH23.3	39918	4038	3542	4151	3414	2957	2164	1890	2252	2928	4003	4429	4149	(12)	
(13)	CDH26.7	19423	2043	1766	2112	1612	1275	848	681	940	1433	2118	2425	2172	(13)	
(14)	Wind	WSAvg	4.0	3.7	3.6	3.4	3.6	3.9	4.3	4.5	4.7	4.8	4.6	4.0	3.6	(14)
(15) Precipitation	PrecAvg	463	90	80	93	59	14	7	5	2	2	13	48	68	(15)	
	PrecMax	1010	429	370	370	296	105	50	30	20	21	127	196	465	(16)	
	PrecMin	140	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	193	91	77	85	61	22	11	7	4	5	27	48	82	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	37.0	37.2	37.0	35.2	34.2	33.2	34.2	37.0	37.2	38.1	38.1	(19)	
		MCWB	21.6	21.7	21.8	21.3	21.2	21.0	20.0	20.1	20.6	21.2	21.6	21.2	(20)	
	2%	DB	36.0	36.0	36.2	35.8	34.1	33.1	32.1	33.1	35.2	36.1	37.1	36.9	(21)	
		MCWB	21.2	21.7	21.7	21.5	21.5	20.6	20.1	19.7	20.3	20.9	21.7	21.5	(22)	
	5%	DB	35.1	35.0	35.2	34.2	33.1	32.1	31.1	32.1	34.1	35.1	36.1	35.9	(23)	
		MCWB	21.3	21.9	22.1	22.2	21.6	20.7	19.9	19.8	20.1	20.7	21.4	21.3	(24)	
	10%	DB	34.1	34.0	34.2	33.1	32.1	31.0	30.1	31.1	33.0	34.1	35.1	34.8	(25)	
		MCWB	21.4	22.0	22.3	22.2	21.8	20.6	19.9	19.4	19.9	20.6	21.4	21.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	25.1	25.2	25.2	24.6	23.2	23.0	22.2	22.8	23.7	24.6	25.2	(27)	
		MCDB	28.4	30.9	30.7	30.2	29.8	29.9	27.9	29.1	31.7	30.5	29.8	29.7	(28)	
	2%	WB	24.6	24.5	24.6	24.5	23.7	22.7	22.2	21.6	21.9	23.1	23.8	24.2	(29)	
		MCDB	28.2	30.2	30.6	29.7	29.1	28.9	27.5	28.6	30.0	30.9	31.2	29.6	(30)	
	5%	WB	23.9	23.9	24.1	23.9	23.3	22.2	21.6	21.1	21.4	22.5	23.3	23.6	(31)	
		MCDB	28.1	29.5	30.3	29.3	29.0	28.0	27.3	28.3	29.8	30.5	31.2	29.6	(32)	
	10%	WB	23.1	23.5	23.6	23.5	22.9	21.7	21.1	20.5	21.0	21.9	22.7	23.1	(33)	
		MCDB	29.5	29.4	30.0	29.1	28.7	27.5	27.1	27.7	29.6	29.9	31.1	29.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.4	10.7	10.0	9.8	10.1	10.3	11.4	11.9	11.7	11.4	10.7	(35)	
		MCDBR	11.6	11.5	11.8	11.4	11.2	11.6	11.3	12.3	13.1	12.4	12.3	12.2	(36)	
		MCWBR	3.3	3.1	3.3	3.2	3.1	3.3	2.9	3.3	3.7	3.4	3.8	3.8	(37)	
	5% WB	MCDBR	9.5	9.7	10.0	9.3	9.3	9.8	9.6	11.2	11.7	11.0	10.8	9.8	(38)	
		MCWBR	3.0	3.0	3.0	2.9	2.9	3.2	3.0	3.5	3.6	3.5	3.5	3.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.382	0.383	0.372	0.354	0.347	0.337	0.352	0.378	0.407	0.407	0.404	(40)	
	taud		2.472	2.506	2.506	2.528	2.591	2.585	2.607	2.508	2.416	2.371	2.393	2.413	(41)	
	Ebn at Noon		951	953	936	918	907	898	915	925	927	918	927	935	(42)	
	Edh at Noon		118	114	112	104	93	91	91	105	120	129	127	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.26	5.99	5.88	5.48	4.96	4.78	5.04	5.88	6.57	6.71	6.52	6.24	(44)	
	RadStd		0.53	0.39	0.33	0.35	0.35	0.28	0.24	0.22	0.29	0.35	0.41	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page